Water Body Detection in High-Resolution SAR Images With Cascaded Fully-Convolutional Network and Variable Focal Loss

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***Abstract*— The water body detection in high-resolution synthetic aperture radar (SAR) images is a challenging task due to the changing interference caused by multiple imaging conditions and complex land backgrounds. Inspired by the excellent adaptability of deep neural networks (DNNs) and the structured modeling capabilities of probabilistic graphical models, the cascaded fully-convolutional network (CFCN) is proposed to improve the performance of water body detection in high-resolution SAR images. First, for the resolution loss caused by convolutions with large stride in traditional convolutional neural network (CNN), the fully-convolutional upsampling pyramid networks (UPNs) are proposed to suppress this loss and realize pixel-wise water body detection. Then considering blurred water boundary, the fully-convolutional conditional random fields (FC-CRFs) are introduced to UPNs, which reduce computational complexity and lead to the automatic learning of Gaussian kernels in CRFs and the higher boundary accuracy. Furthermore, to eliminate the inefficient training caused by imbalanced categorical distribution in the training data set, a novel variable focal loss (VFL) function is proposed, which replaces the constant weighting factor of focal loss with the frequency-dependent factor. The proposed methods can not only improve the pixel accuracy and boundary accuracy but also perform well in detection robustness and speed. Results of GaoFen-3 SAR images are presented to validate the proposed approaches.**

***Index Terms*— Conditional random field (CRF), fully convolutional network (FCN), high-resolution synthetic aperture radar (SAR), loss function, water body detection.**

I. Introduction

With the advantages of all time, all-weather and high resolution, synthetic aperture radar (SAR) is widely used in agriculture management, target detection, disaster monitoring, and other fields [1]–[5]. Research on water body detection in SAR images takes on great significance in flood monitoring, comprehensive investigation, and environmental resource planning [6]–[9]. Especially, the result of water body detection strongly influences the performance of ship detection in SAR images [10]. In recent years, with the development of radar devices and related satellite technologies, it is possible to obtain a large number of high-resolution and large-scene spaceborne SAR images quickly and conveniently. It has been a critical problem on how to extract water areas from these images automatically. Unlike multiclass SAR image classification algorithms that try to interpret the image's semantic information intuitively [11]–[13], water body detection is a two-class pixel-by-pixel classification that focuses more on higher pixel accuracy and more explicit segmentation boundaries [7], [8]. More specifically, water body detection algorithms classify the water bodies such as rivers, lakes, and oceans in SAR images as attractive regions, while regarding other land scenes as background regions. However, as shown in Fig. 1, the performance of water body detection in high-resolution SAR images is easily affected by complex land regions due to their sharing the characteristics of different continuities and low scattering intensity with water areas. Meanwhile, the differences between different water bodies, e.g., the isolated ponds or lakes usually reflect low scattering energy and the oceans with large waves reflect high and diverse scattering energy, could affect the detection results seriously. Due to this complex interference, it is hard to realize the robust and automatic water body detection in high-resolution SAR images.

A variety of algorithms have been proposed aiming at the water body detection in SAR images, which can be categorized into two kinds: unsupervised learning and supervised learning. The unsupervised-based detection methods classify an image with amplitude or extracted features and do not need any annotation results or additional information. These methods can be summarized as follows: threshold methods [6], [14], clustering methods [15], Markov random field (MRF) methods [16], and edge detection methods [17]. These methods perform well accuracy and efficiency in the detection of low-resolution SAR images with plain scene. However, as for high-resolution SAR images with more geometric and radiometric details [11], the amplitude variation caused by complicated background and various water bodies results in detection performance sliding

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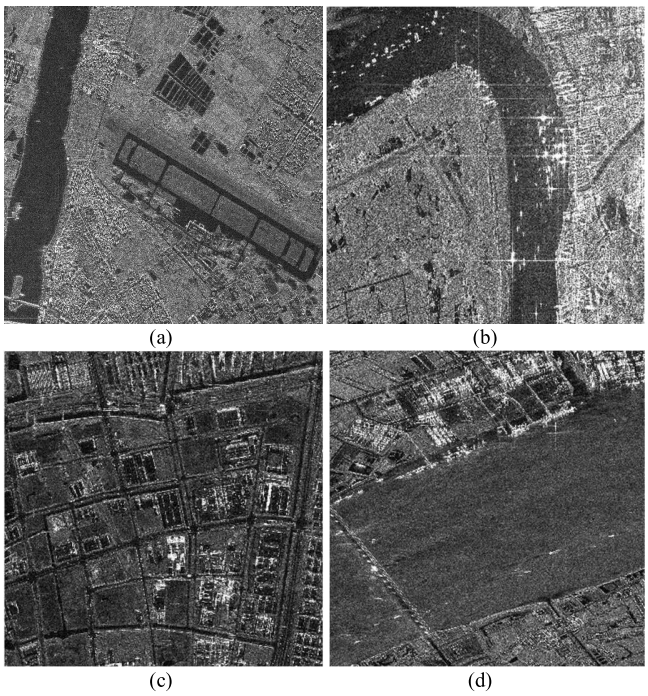

Fig. 1. Few patches of GaoFen-3 images. Some land areas in these patches are easily misclassified as water areas by different detection algorithms. (a) Airport region is easy to classified as water areas. (b) Strong scattering energy of ship targets makes their surrounding areas highly to be classified as land areas. (c) Highways and some shadows of buildings are easy to classified as water areas. (d) River areas with strong scattering energy caused by powerful waves and winds are easy to classified as land areas.

down significantly. The reason mainly lies in that they cannot make use of contextual information or semantic knowledge (e.g., the low-amplitude area nearby the buildings is more likely to be land area, and the region with continuous and low-amplitude pixels is more likely to be water area).

To improve detection accuracy in high-resolution SAR images, a number of supervised-based (or data-driven) methods have been proposed. The combination of excellent feature extraction and robust classifier is much essential for the water body detection of SAR images. As for feature extraction, the gray-level co-occurrence matrix (GLCM) based on raw pixels is one of the most typical techniques [18], which makes use of the information of mean, variance, and contrast. Transform domain methods are also widely used in the feature extraction of SAR images, which can be categorized into follows: 2-D Flourier coefficients, wavelet transform, Gabor wavelet transform [19]. As for classifiers, the support vector machine (SVM) [20], [21], logistic regression (LR), sparse representation classifier (SRC) [22], and random forest (RF) [23] have been adopted with labeled samples. However, handcrafted filters in these methods need to be determined by prior knowledge and experience, which tend to be difficult and time consuming for the water-body detection in high-resolution SAR images [13].

Recently, owing to its excellent representational capacities, deep neural network (DNN) has become widely used in computer vision [24]–[27]. One of the most successful DNNs is the convolutional neural network (CNN), which was first proposed for the image classification of the Mixed National Institute of Standards and Technology database (MNIST) data set [26]. Composed of convolutional layers, pooling layers, and activation layers, a well-trained CNN can extract more abstract and powerful features from the testing data set and output the most likely classification results. What is more, the introduction of new layers like region proposal network (RPN) and deconvolutional layers makes CNN a powerful method for object detection and semantic segmentation [28]. Inspired by the successful application to natural and optical image processing, CNNs are also used in the interpretation of remote sensing images.

A. Related Work

The water body detection in SAR images is a binary semantic segmentation in the field of computer vision or a two-class image classification in remote sensing [7], [8]. CNNs have first achieved massive success for optical and hyperspectral remote sensing image classification [9], [29]–[32]. Marmanis *et al.* [33] trained the CNN using ImageNet pertained networks. Cheng *et al.* [34] proposed a simple and effective method to learn discriminative CNNs to boost the performance of optical remote sensing image classification. Hu *et al.* [30] employed a typical CNN structure to classify images in the spectral domain. Chen *et al.* [29] proposed a regularized 3-D CNN-based feature extraction model and a hybrid framework for the hyperspectral data. Different from those popular methods for optical or spectral image classification, the lack of annotated data sets results in that the use of CNNs over SAR image classification is limited [35]. Zhou *et al.* [12] innovatively converted polarimetric SAR data into a normalized 6-D real feature vector and proposed a four-layer CNN for image classification. Geng *et al.* [11], [13] introduced a deep recurrent encoding neural network and deep convolutional autoencoders for high-resolution SAR image classification, which achieve the highest classification accuracy in testing images. Although these CNN-based methods perform very high accuracy, the convolutions with large stride and the sliding window on large-scene images could cause a few contradictions between high detection accuracy and fast running speed [7].

To solve the above issues, the fully-convolutional network (FCN) and its variants have been proposed [24], [36], [37]. Maggiori *et al.* [37] first proposed an end-to-end FCN for the classification of large-scale remote-sensing images. For the water-body detection from a high-resolution satellite image, Miao *et al.* [8] proposed a restricted receptive field deconvolution network (RRF DeconvNet) and edges weighting loss (EWLoss). In order to use the contextual information of raw pixels, Feng *et al.* [7] introduced fully connected conditional random field (FullCRF) into FCNs and obtained more explicit water body boundaries, while the complex message passing and manual hyperparameter optimization in the FullCRF result in a long-running time [25] and unsatisfactory detection accuracy. Shahzad *et al.* [35] added a CRF represented as recurrent neural network (RNN) into the FCN structure, which realizes the automatic building detection in VHR SAR images. It is worth noting that although the introduction of RNN structure can significantly improve the detection accuracy, there is

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no guarantee that the extracted boundary of buildings is as clear as the ground truth [35]. As a result, if these available networks are directly applied to the water body detection of high-resolution SAR images, blurred water boundaries and visual degeneration will still exist.

B. Contributions

As shown above, there are two critical issues in the water body detection of high-resolution SAR images: one is how to detect most of the water areas from input SAR images with low false alarm rate, another is how to ensure clear water body boundaries based on high detection accuracy. For the first question, the upsampling pyramid networks (UPNs) are proposed, which first extract high-dimensional features from input image patches, then establish a decoder network for the pixel-wise detection result. For the second question, the fully-convolutional CRFs (FC-CRFs) take the probability maps generated by UPNs and the original pixel values of patches as input and output the final water body detection results. Therefore, as shown in Fig. 2, the proposed deep networks consist of two major parts: UPNs and FC-CRFs, and the whole end-to-end detection framework are named as cascaded FCN (CFCN). The major contributions of this article are indicated as follows.

1) Proposed UPNs can extract more powerful and semantic features from input SAR images with the combination of the Xception network and atrous spatial pooling pyramid (ASPP) network. What is more, the decoder network of UPNs is established based on multiple upsampling layers, which can make use of multi-dimensional features and generate probability distributions for each pixel in each SAR image.
2) FC-CRFs are proposed to suppress the resolution loss caused by UPNs and preserve clear water boundaries, which takes probability maps generated by UPNs and raw pixels of input images as input. It is noting that FC-CRFs reformulate all key steps in the mean-field approximation of traditional CRFs as convolution layers, which can not only run an inference quickly on graphic processing units (GPUs), but also realize the automatic learning of all hyperparameters.
3) A novel variable focal loss (VFL) function is proposed to eliminate the inefficient training of positive/negative samples or hard/easy samples caused by imbalanced category distribution, which was ignored in most published articles. Specifically, the introduction of the weighting factor and regulating factor offers VFL the capacity to adjust the loss contribution based on the differences of input SAR images, further makes the network mine more hidden information of these images in the training process and improves detection accuracy.

The rest of this article is organized as follows. The background of the FCN is reviewed in Section II. Section III first introduces the UPNs, then presents the FC-CRFs, further proposes the novel VFL function. The experimental results in high-resolution SAR images are analyzed in Section IV. Finally, the conclusions are drawn in Section V.

II. Background on FCN

In machine learning, an artificial neural network is a system of interconnected neurons that pass messages to each other [37]. As a typical network structure in computer vision, CNN performs well feature representation ability, for it combines a number of layers, e.g., convolutional layer, pooling layer, activation layer, and fully-connected layer. However, with the densely connected spatial relationship and independent weights between different neurons, the quantity of weight parameters of fully connected layers far exceeds that of convolutional layers, which increases the risk of over-fitting when training a network [36], [38]. Replaced all fully-connected layers of traditional CNNs with convolutional layer, FCN performs very well in dense pixel-wise classification. Two of the most typical network structures in FCN are introduced in this section.

A. Feature Extraction Network

The vital advantage of FCNs is that they can extract powerful and contextual features from input images. There is a large number of FCN-based feature extraction networks, e.g., VGGnet, ResNet, DenseNet, and Xception [25], [39]. Taking these FCNs as backbone, one can conveniently establish target classification networks, object detection networks, and semantic segmentation networks. What is more, initialized with the well-trained weights based on these FCNs, the established network performs better training stability and faster convergence speed than that of random initialization [33], [38]. How to extract more robust features with fewer parameters has been a critical issue when devising a novel FCN structure. Based on the hypothesis that the mapping of cross-channel correlations and spatial correlations in the feature maps of CNNs can be entirely decoupled, Xception is proposed as a linear stack of depthwise separable convolution layers with residual connections [25]. It is noticeable that the depthwise separable convolution decouples the conventional convolution into depthwise convolution and pointwise convolution, which can decrease the quantity of parameter weights significantly.

As Fig. 3 shows, Xception is composed of three flows: the entry flow, the middle flow, and the exit flow. The entry flow contains three cascaded residual modules [40]. Taking the first module as an example, this module takes the original image as input, on the one hand, it uses three cascaded depthwise separable convolutions to extract high-dimensional features, on the other hand, it uses a conventional convolution to map the original image features to the generated feature maps. The middle flow in Fig. 3(b) uses 16 repetitive subflows to make more nonlinear transformations of the feature maps of entry flow. With fixed convolution stride of one pixel and a fixed number of convolution kernels, the size and the number of feature maps remain unchanged. The network structure of the exit flow is shown in Fig. 3(c). Flowing through the Xception network, the original input images are converted into 2048 collateral feature maps which size is one-sixteenth of the original image.

As discussed above, once the FCNs are well trained, powerful, and contextual features can be extracted from input images

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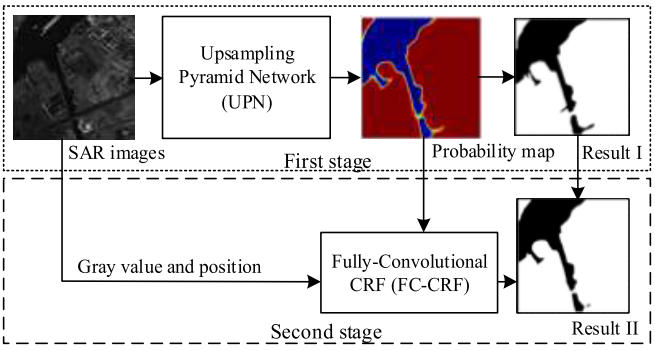

Fig. 2. Overview of the proposed CFCN.

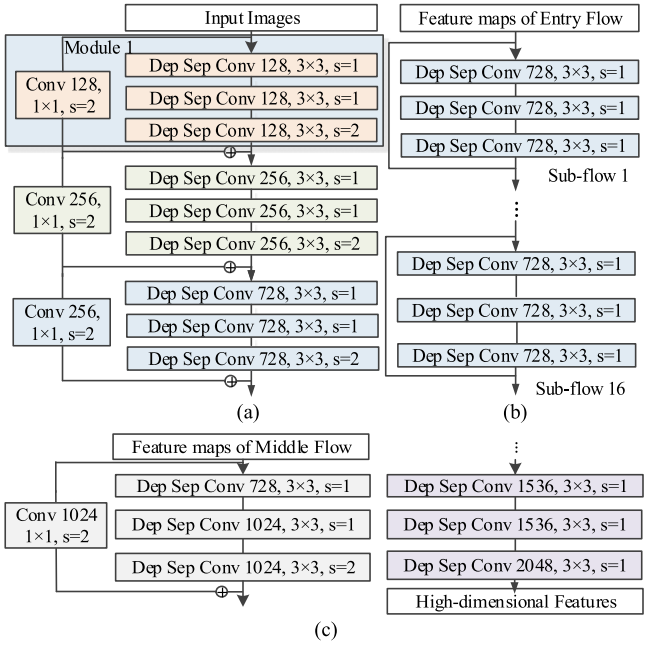

Fig. 3. Xception network for extracting high-dimensional features of input images. "Dep Sep Conv 128, 3 × 3, $s = 1$" means that uses 128 depthwise separable convolution kernels with size of 3 × 3 pixels and stride of 1 × 1 pixel. (a) Entry flow. (b) Middle flow. (c) Exit flow.

by repetitive convolution layers. Thus, the obtained feature maps can be used to establish a more complicated network for image detection.

B. Upsampling Layer Based on Interpolation

To attain a pixel-to-pixel detection result corresponding to the input image, feature maps with low resolution should be upsampled to a higher resolution. After the success of the network visualization, the deconvolution layer (also named as "transposed convolution" or "fractional convolution") became widely adopted in the context of semantic segmentation [7], [24], [36]. The size of convolution kernels can be represented as (o, i, k_w, k_h) which means output channels, input channels, kernel width, and kernel height. A simplified deconvolution is shown in Fig. 4. For an input feature map with (1,4,4) and a kernel with (1,1,3,3), the input feature map is first converted into a fractional feature map with (1,9,9), where the yellow pixels are the original

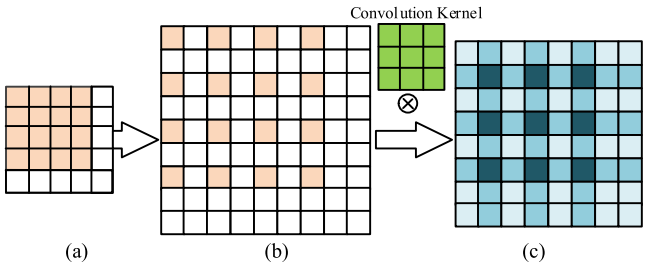

Fig. 4. Deconvolution layer. (a) Input feature map. (b) Fractional feature map. (c) Output feature map.

input pixels, and the white ones are the zero-padded pixels. Then the kernel with (1,1,3,3) is convolved with the fractional feature map, which generates a feature map with (1,8,8). It is noticeable that the "uneven overlap" in deconvolution causes a strange checkerboard pattern of artifacts in the output feature map [37]. In particular, when the kernel size is not divisible by the stride (the spacing between zeros padded pixels), uneven overlap could happen, and it will lead to visual degeneration in the generated feature map. While the network could learn unequal weights to avoid this, neural network struggle to avoid it completely [24].

To upsample feature maps without the above artifacts, a deconvolution layer is divided into an image-resizing layer and a convolution layer. To be specific, the input feature map $\mathrm{X}^{\mathrm{inp}}$ is first amplified to feature map $\mathrm{X}^{\mathrm{amp}}$ with a bilinear interpolation, which is given by

$$\mathrm{X}^{\mathrm{amp}}_{(i,j)} = \sum_{\alpha,\beta=0}^{1} |1-\alpha-\{i/\lambda\}||1-\beta-\{j/\lambda\}|\mathrm{X}^{\mathrm{inp}}_{(m,n)} \quad (1)$$

where $m = [i/\lambda] + \alpha n = [j/\lambda] + \beta\lambda$ is an upsampling factor, which equals the value of the size of the output feature map divided by that of input feature map $[\cdot]$ and $\{\cdot\}$ is the integral and fractional part, respectively. Subsequently, the output feature map $\mathrm{X}^{\mathrm{out}}$ can be generated by convolving $\mathrm{X}^{\mathrm{amp}}$ and kernel with size $(o, i, 1, 1)$. It is worth noting that 1 × 1 convolution does not change feature map size and only transforms channel features. Based on the combination of resizing layer and 1 × 1 convolution layer, feature maps with low resolution can be upsampled to a higher resolution without visual degradation [24], [37].

III. Methodology

A. Step 1: UPN

Based on the above feature extraction network and upsampling layer, a UPN is proposed for the water body detection in high-resolution SAR images. As Fig. 5 shows, this network consists of two structures: the encoder network for extracting high-dimensional features of input SAR images and the decoder network for giving the probability distribution of detection result.

1) Encoder Network Based on A Multi-Scale Architecture: The encoder network is composed of a feature extraction network and an atrous spatial pyramid pooling (ASPP) module [24]. Flowing through the above Xception network, the size of feature maps reduces to one-sixteenth of the original

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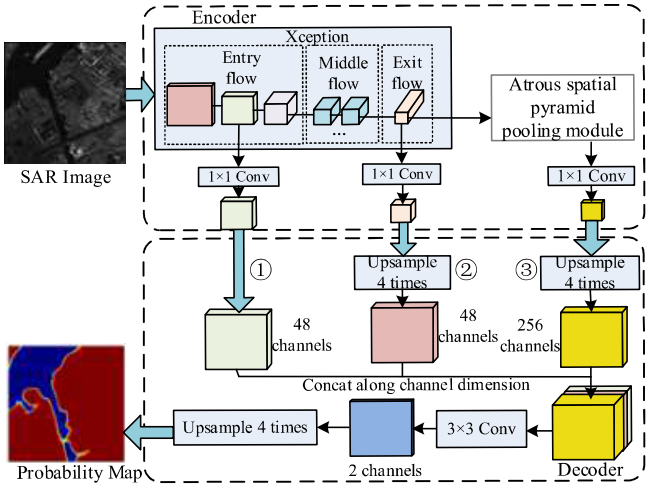

Fig. 5. Structure of UPN.

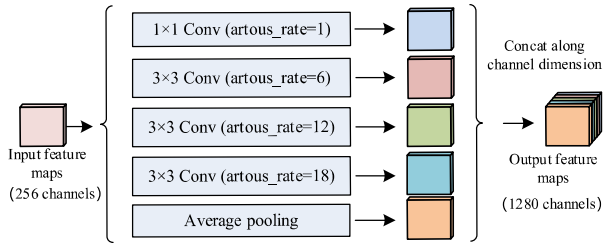

Fig. 6. Structure of atrous spatial pyramid network.

input image, while the number of feature maps increases to 2048. Meanwhile, the ASPP module pays more attention to capturing multiscale features by dilated convolution [24]. As shown in Fig. 6, dilated convolutions with different atrous rates are used to get three groups of feature maps with various receptive fields, while 1×1 convolution and global average pooling are used to preserve the original features of input maps. Finally, the encoder network merges each feature map of the ASPP module along the channel dimension and uses the 1×1 convolution to reduce the size of combined feature maps.

2) Decoder Network Based on Multiple Upsampling Layers: In order to extract more semantic information of feature maps and reserve the resolution information of input image, the decoder network takes the feature maps of the second convolution module of the entry flow as the input of low-dimensional feature (① in Fig. 5), the output feature maps of Xception as the middle-dimensional feature (② in Fig. 5) and the output feature maps of the encoder network as the input of high-dimensional feature (③ in Fig. 5), respectively. Based on the above upsampling layers with bilinear interpolation, these three feature mapping networks form a UPN that can capture multidimensional features of original SAR images. After a few convolution layer and upsampling layer, the UPN could generate a probability map which represents the probability distribution to each class of the input image. Subsequently, the category of each pixel can be determined approximately by taking the maximum along the channel dimension. Avoiding the artifacts of transposed convolution, the feature maps in UPN are upsampled by the combination of bilinear interpolation and 1×1 convolution layer.

3) Detailed Networks Settings of Proposed UPNs: Except for the skip connection layer in Xception, all convolution layers in UPNs are followed by the nonlinear layer and batch normalization layer. The nonlinear layer uses the rectified linear unit (ReLU) function to transform the feature maps. Considering a mini-batch $\mathcal{B}$ of size m, the batch normalization layer can be expressed as [41]

$$\hat{x}_i \leftarrow \frac{x_i - \mu_B}{\sqrt{\sigma_B^2 + \varepsilon}} \tag{2}$$

where x_i is the activation of ith sample, μ_B and σ_B^2 are the mean and variance of the mini-batch, ε is a constant for numerical stability, and $\hat{x}_i$ is the normalized activation. To make sure that the inserted transformation can represent the identity transform, a pair of parameters γ and β is introduced to scale and shift the normalized activation $\hat{x}_i$

$$y_i = \gamma \hat{x}_i + \beta \tag{3}$$

where y_i is a linear transformation of $\hat{x}_i$.

To improve the generalization ability of this detection network and prevent overfitting, this article also introduces 2-D dropout [26] to UPNs. Thinking $x^{(l)} \in \mathbb{R}^{m,n}$ as a feature map in the final 1×1 convolution layer of the encoder network, during once training, the dropout layer will first generate a corresponding coefficient matrix $\mathrm{r}^{(l)} \in \mathbb{R}^{m,n}$. Each element in this matrix belongs to the distribution

$$\mathrm{r}_{i,j}^{(l)} \sim \text{Bernoulli(p)} \tag{4}$$

where p is the activation probability which needs to be set manually. Then, the layer uses the coefficient matrix $\mathrm{r}^{(l)}$ to adjust the activation value of $x^{(l)}$

$$\hat{x}^{(l)} = \mathrm{r}^{(l)} * x^{(l)}. \tag{5}$$

The coefficient matrix is regenerated during each training. When testing the segmentation model, the feature map $x^{(l)}$ should be multiplied with the probability p.

B. Step 2: Fully Convolutional CRF

The convolution layers with large stride in FCNs lead to the resolution loss of original input images and blurred boundaries between water areas and land areas [7], [8]. To alleviate this type of loss, a number of articles [7], [24], [35], [42], [43] try to introduce the CRF into FCNs, which can make use of structural information of input images. However, with the nontrivial problem of permutohedral lattice [43], [44] in Message Passing, the variance parameters in Gaussian kernels of FullCRFs are still optimized manually, which severely affects the detection accuracy. Meanwhile, as shown in Fig. 7(a), the fully-connected pairwise potential of FullCRFs leads to a huge computation burdon. Therefore, a novel locally-connected FC-CRF is introduced in this section.

As shown in Fig. 7(b), the FC-CRF supplements FullCRFs [24], [45] with a conditional independent assumption that when the Manhattan distance d holds $\mathrm{d}(i_x, j_x) > k$ or $\mathrm{d}(i_y, j_y) > k$, the label distribution of two pixels (i_x, i_y), (j_x, j_y) are conditionally independent. It means that the pairwise energy $\varphi_{\mathrm{p}}(\cdot)$[22] is zero for all pixel pairs (i, j)

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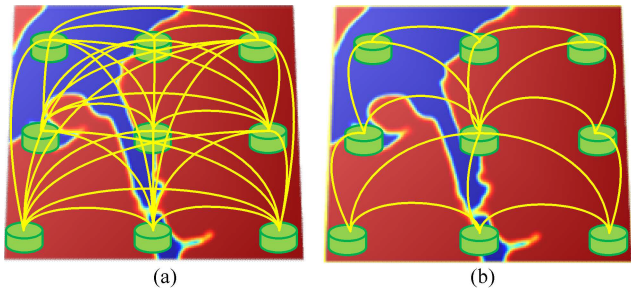

Fig. 7. Different connected structures of pairwise potential between full CRFs and FC-CRF. (a) Pairwise potential of each other pixels could affect the label distribution of current pixel as for FullCRFs. (b) Label distribution of current pixel is only affected by the pairwise potential of its neighborhood pixels as for FC-CRFs.

whose distance exceeds k the input image. Therefore, the size of the receptive field of Gaussian kernels in FC-CRF is practically the Manhattan distance k. Based on this assumption, the message passing in mean-field approximation [45] is converted into a convolutional layer in CNN, which means that all critical steps in mean-field approximation can be reformulated as standard CNN structures [43]. The structure of FC-CRF and corresponding inference method is shown in Fig. 8. The algorithm changes are mainly reflected in the following.

1) Message Passing With Variance-Sharing Convolution: FC-CRF reformulates the intricate permutohedral lattice [24], [45] of FullCRF as a convolutional layer based on a conditional independent assumption. Assuming δ_i and δ_j as offsets relative to the center coordinate (i, j) of Gaussian kernels, where $\delta_i, \delta_j \in \{-k-12, -(k-1/2)+1, \ldots, +(k-1/2)\}$.

A convolutional kernel in FC-CRF can be formed as

$$k_g\left[\delta_i, \delta_j, i, j\right] = \exp\left(-\sum_{\mathrm{d}=1}^{D} \frac{\left|f_{\mathrm{d}}[i, j] - f_{\mathrm{d}}\left[i-\delta_i, j-\delta_j\right]\right|^2}{2\theta_{\mathrm{d}}^2}\right) \tag{6}$$

where D is the number of feature vector, for the appearance kernel $D = 2$ and for smoothness kernel $D = 1$. f_{d} is the position or the gray value of input image. As shown in Fig. 8, the Gaussian kernels matrix K with dimension (k, k, w, h) can be attained by substituting (6) into the appearance kernel and smoothness kernel of FullCRF [45], where (w, h) is the size of the image I. As a result, for each $k \times k$ convolution kernel in K, weight factors are not initialized and trained randomly, but are constrained by the variance θ_{d} of Gaussian function and feature vectors $\mathbf{f}$ of the input image I. Inspired by the definition of convolution in CNN, this particular convolution is named as variance-sharing convolution. It is noticeable that this convolution is a specific form of locally connected convolution.

The forward propagation of variance-sharing convolution can be efficiently implemented on GPUs. During training, we need to backpropagate the gradient of loss l to the parameters of this convolution [46]. The back-propagation gradient of variance-sharing convolution with chain rule is as follows:

$$\frac{\partial l}{\partial \theta_{\mathrm{d}}} = \frac{\partial l}{\partial k_g} \cdot \frac{\left|f_{\mathrm{d}}[i, j] - f_{\mathrm{d}}\left[i-\delta_i, j-\delta_j\right]\right|^2}{\theta_{\mathrm{d}}^3} \cdot \exp\left(-\sum_{\mathrm{d}=1}^{D} \frac{\left|f_{\mathrm{d}}[i, j] - f_{\mathrm{d}}\left[i-\delta_i, j-\delta_j\right]\right|^2}{2\theta_{\mathrm{d}}^2}\right). \tag{7}$$

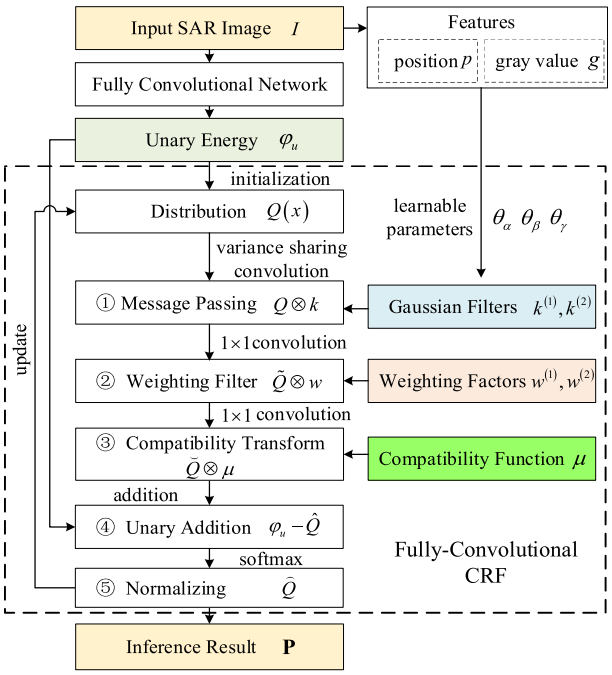

Fig. 8. Structure of FC-CRF and corresponding inference method.

2) Weighting Filter and Compatibility Transform: As shown in Fig. 8, the weighting filter in mean-field approximation can be reformulated as the 1×1 convolution with L input channels and one output channel. As for the compatibility transform, it can share the outputs from the weighting filter $\breve{Q}$ between different labels to a different extent. The common used Potts model parameterizes compatibility function in [43] and [45] as $\mu(l_1, l_2) = [l_1 \neq l_2]$. However, this model cannot balance the penalty quantitatively between various labels. Therefore, this article reformulates this compatibility transform as 1×1 convolution with L input channels and L output channels, which leads to more freedom of compatibility function.

3) Unary Addition and Normalizing: The unary addition in Fig. 8 is a pure forward-propagation and backpropagation of the CNN structure. The normalizing step $Q_i \leftarrow 1/Z_i \exp\left\{\widehat{Q}_i(l)\right\}$ can be viewed as a softmax transform for normalizing the distribution function $\widehat{Q}$ of all labels [43].

With the above vital transformations, a FullCRF is reformulated as an FC-CRF. Therefore, the traditional learning methods of FullCRFs, e.g., iterative scaling and stochastic gradient descent, can be replaced with the training methods of DNNs, which realizes the automatic learning of all hyperparameters in this model, especially the variances of Gaussian kernels. Since all steps in FC-CRF has been reformulated as convolutional layers, it is actually a particular type of FCN which makes

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use of the resolution information of input SAR images and probability distribution generated by CNNs.

C. VFL for Imbalanced Distribution

As for an SAR image classification data set, it is challenging to make sure the same number of pixels in each category for that the landmark distribution of high-resolution SAR images varies widely [11]–[13], [20], [47]. Thus, most SAR classification data sets in CNNs-based classification methods are imbalanced data sets. These imbalances can lead to a few easily-neglected problems in the training of networks.

1) Cross-Entropy Loss Function in CNNs: Assuming an image I with corresponding ground truth X, for each pixel i in I, $X_i \in \{0, 1\}$ is the ground truth of the pixel i, and $P_i \in [0, 1]$ is the corresponding predicted value. The cross-entropy loss [36] of the pixel i is

$$\mathrm{CE}(P_i, X_i) = -X_i \log(P_i) - (1 - X_i) \log(1 - P_i). \quad (8)$$

For notational convince, we define P_i^t as

$$P_i^t = \begin{cases} P_i, & X_i = 1 \\ 1 - P_i, & X_i = 0 \end{cases} \quad (9)$$

and rewrite $\mathrm{CE}(P_i, X_i) = \mathrm{CE}\left(P_i^t\right) = -\log\left(P_i^t\right)$.

Assuming h and w as the height and width of I, respectively. The total loss of all pixels in I follows:

$$\begin{aligned} L_{\mathrm{CE}}(P) &= \frac{1}{w \cdot h} \sum_{i=1}^{h} \sum_{j=1}^{w} \mathrm{CE}\left(P_{i,j}^t\right) \\ &= -\frac{1}{w \cdot h} \sum_{i=1}^{h} \sum_{j=1}^{w} (X_i \log(P_i) + (1 - X_i) \log(1 - P_i)). \end{aligned} \quad (10)$$

Thus the cross-entropy loss of each pixel to the total loss equals to each other, which means that when the quantity of background pixels ($X_i = 0$) is much larger than that of positive pixels ($X_i = 1$), the training of detection network is inefficient as it is dominated by these background pixels. In detail, the above imbalances could cause two problems [48]: 1) the training is ineffective as the contribution of inference loss of positive pixels (water regions) and background pixels (land regions) to the total inference loss is inconsistent and 2) the significant characteristic differences between the areas belonging to the same category, like the backscattering energy difference of open seas and landlocked lakes, are ignored, lead to the degeneration of detection networks. To alleviate these imbalances, a novel VFL is proposed in follows.

2) Novel VFL Function: Focal loss, a type of loss function commonly used in target detection [48], considers the severe imbalance between easy and hard samples as well as between positive and negative samples. The original Focal loss can be expressed as

$$L_{\mathrm{FL}}(P) = -\alpha^t \left(1 - P^t\right)^{\gamma} \log\left(P^t\right) \quad (11)$$

where $\alpha^t \in [0, 1]$ is a weighting factor, similar to P^t. In target detection, α^t is a manual hyper-parameter which is trying to weaken the imbalance between the positive and negative samples. The regulating factor $\left(1 - P^t\right)^{\gamma}$ is introduced to weaken the imbalance between easy and hard samples, where $\gamma \geq 0$ is a tunable focusing parameter, running through a positive number.

If this focal loss is directly extended to the water body detection network, the loss should be written as

$$L_{\mathrm{FL}}(P) = -\frac{1}{w \cdot h} \sum_{i=1}^{h} \sum_{j=1}^{w} \alpha_{i,j}^t \left(1 - P_{i,j}^t\right)^{\gamma} \log\left(P_{i,j}^t\right) \quad (12)$$

where $\alpha_{i,j}^t$ is a weighting factor similar to α^t in (11) and related to the pixel (i, j) of the image I. The larger the image I is, the more pixels it contains, leading to the difficulty of determining a vast number of $\alpha_{i,j}^t$ parameters manually. Meanwhile, $\alpha_{i,j}^t$ should be related to the water pixel frequency of image I: if the water frequency is too high, the weighting of the background pixels should be increased massively to balance the learning efficiency of the two classes, and vice versa. For solving these problems, the VFL is proposed as follows:

$$L_{\mathrm{VFL}}(P) = -\frac{1}{w \cdot h \cdot m} \sum_{k=1}^{m} \sum_{i=1}^{h} \sum_{j=1}^{w} \alpha_k^t \left(1 - P_{i,j}^t\right)^{\gamma} \log\left(P_{i,j}^t\right) \quad (13)$$

where m is the size of mini-batch when training a network. The α_k^t in (13) is

$$\alpha_k^t = \begin{cases} \alpha_k, & X_{i,j} = 1 \\ 1 - \alpha_k, & X_{i,j} = 0. \end{cases} \quad (14)$$

Assuming the emergence frequency of water pixels in the image I_k as f_k, α_k is inversely proportional to f_k

$$\alpha_k = 1/f_k. \quad (15)$$

To avoid unstable training caused by too high or too low α_k, we can manually set the minimum and the maximum of α_k as $\alpha_{\min}$, $\alpha_{\max}$, respectively. Thus, the new VFL function makes use of the weighting factor α_k^t to balances the learning efficiency of the positive and background pixels. Meanwhile, the modulating factor $\left(1 - P^t\right)^{\gamma}$ is utilized to mitigate the effect of the easy-classified pixels on the total loss, and extend the range in which a pixel receives a low loss. For example, when $\gamma = 2$ a pixel classified with $P^t = 0.9$ would have $100\times$ lower loss compared with cross-entropy loss and with, $P^t = 0.968$ it would have a nearly $1000\times$ lower loss [48]. Therefore, VFL prevents the vast number of easy negative pixels from overwhelming the detection network during training.

To intuitively understand the VFL function, a training process on the water body detection training data set is shown in Fig. 9. Since that the weighting factor α can be adaptively adjusted to the water pixels frequency f, the VFL performs a higher loss value than cross-entropy loss. Although the VFL has an unstable convergence in Fig. 9(b), the experimental results in Section IV validate the effectiveness of the proposed VFL function.

To prevent over-fitting and further improve the generalization ability [8], an L2 regularization term is added to the

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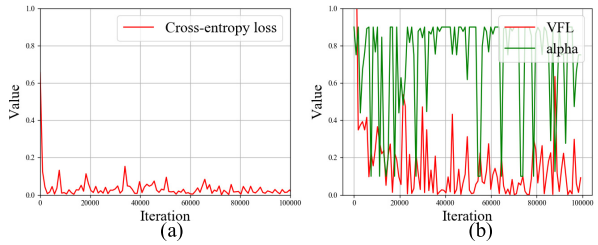

Fig. 9. Variation of cross-entropy loss and VFL with the training iteration. (a) Cross entropy loss can easily be converged to a local minimum value. (b) VFL and weight factor α, $\alpha \in [0.1, 0.9]$. It can be seen that the VFL is affected by α, which is inverse to the water pixels frequency in each training iteration.

VFL function. Supposing θ_i is one of the weight factors of all convolution kernels, the total loss function is

$$L_{\text{total}} = L_{\text{VFL}} + \frac{1}{2}\eta \sum_{i=1}^{m} \theta_i^2 \tag{16}$$

where m is the quantity of all the weight parameters that need to be trained in the detection network and η is the regularization parameter.

IV. Experiments

First, the water body detection data set is described in Section IV-A. Then, the training details are introduced in Section IV-B. The experiments of different detection methods are shown in Section IV-C. Finally, the experiments of multiple-modes images are analyzed in Section IV-D.

A. Water Body Detection Data Set

A data set with large quantity annotated images is much critical for the supervised-based detection method. In some cases, the improvements in detection performance by establishing a complete data set is much more significant than that through creating a novel detection approach [26]. There are few classification data sets of high-resolution SAR images in [11]–[13] and [35]. These SAR images with resolution from 0.38 to 1.5 m, are acquired from Radarsat-2 and TerraSAR-X, et. al. There are different types of objects in these images, like building, road, vegetation, alongside, and water area. However, the water areas cover only a small portion in these images and the quantity of annotated images is not enough for training a very complex detection network. To verify the effectiveness of the proposed network, 15 high-resolution SAR images are utilized in the experiments, which are acquired from the GaoFen-3 satellite, the first C-band multipolarization SAR satellite of China [3]. These images are the second-level data of GaoFen-3 system, and they have been processed by basic SAR processing methods, such as radiometric calibration and geometric rectification. The GaoFen-3 SAR system has 12 imaging modes. The working parameters of five high-resolution imaging modes of them are shown in Table I.

There are different types of water bodies in these images, like great rivers, small streams running parallel to roads, isolated lakes, and paddy fields. The two of these images are shown in Fig. 10(a) and (c). It is a vital issue in determining

TABLE I
Working Parameters With Different Imaging Modes

Imaging Modalities	Resolution (m)	Polarization
Spotlight (SL)	1.0	Single
Ultra-fine stripmap (UFS)	3.0	Single
Fine stripmap I (FSI)	5.0	Dual
Fine stripmap II (FSII)	10.0	Dual
Full polarized stripmap I (QPSI)	8.0	Full

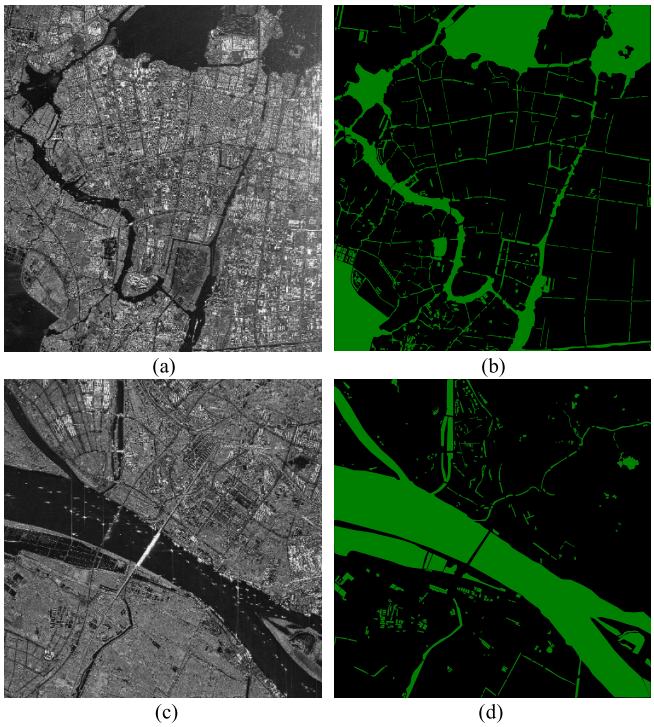

Fig. 10. SL SAR images of GaoFen-3 satellite. (a) Suzhou, Jiangsu Province, China, with size 12 248 × 13 368 pixels and area 163.73 km^2. (b) Ground truth of (a) by manual annotation. (c) Wuhan, Hubei Province, China, with size 11 572 × 9973 pixels and area 115.41 km^2. (d) Ground truth of (c) by manual annotation.

the category of each pixel in these images. We annotate them manually with the help of Google Earth and LabelMe [49], which is an open-source annotation tool for drawing boundaries and transferring format. To be specific, the great rivers are first classified as water areas (or positive samples) according to naked eyes. Then the categories of other small areas are determined according to corresponding optical images in Google Earth. The annotation images or ground truths of Fig. 10(a) and (c) are shown in Fig. 10(b) and (d), respectively. In Fig. 10(b) and (d), the green regions represent the water bodies, and the black areas represent the land regions.

B. Training Details of CFCNs

To train the proposed detection network, 10 out of 15 SAR images are taken as training images. These ten images are acquired on spotlight (SL) mode and HH polarization, and the corresponding azimuth resolution and range resolution are 1.0 m. Since the large size leads to substantial computational load, these images are cropped into the nonoverlapping patches of 513 × 513 pixels. The pixel intensity is rescaled to [0, 255]

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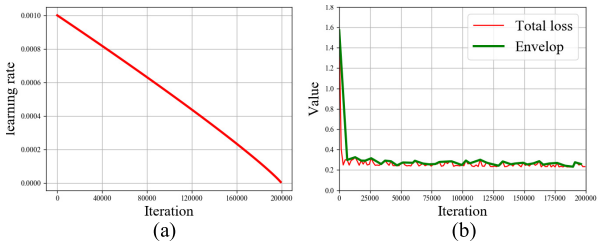

Fig. 11. Variation of learning rate and training loss with the training step. (a) Learning rate. (b) Loss.

by Logarithmic normalization. To prevent overfitting, these patches are augmented by flipping and rotation, as shown in [35] and [47]. Finally, we get 12 796 image patches and corresponding ground truths for training and the remaining 5 out of 15 of the annotated images for testing. Besides the above operations, there are not any augmentation methods for training this detection network.

Avoiding the vanishing gradient [50] and improving the comprehensibility of the proposed detection method, this article realizes the water body detection with the two-stage processing: First, a UPN is trained with VFL function, then the FC-CRF is trained with the probability maps generated by UPN.

1) First Training Stage: In the first training stage, for the loss function in (13), α_k falls into $[0.001, 0.999]$, the tunable focusing parameter γ in (13) is 2, the regularization parameter η is $4e^{-5}$, and the batch size m is 4. The activation probability of the 2-D dropout layer is 0.5. The momentum gradient descent algorithm [46] is used to realize the backpropagation of training loss: the momentum coefficient α is 0.9 and the polynomial algorithm decides the learning rate

$$\varepsilon_i = \varepsilon_0 \cdot \left(1 - \frac{\lambda_i}{\lambda_{\text{global}}}\right)^{\beta} \tag{17}$$

where $\varepsilon_0 = 7e^{-3}$ represents the initial learning rate, $\lambda_{\text{global}} = 2e^5$ is the global training step, and $\beta = 0.9$ is used to adjust the decay of learning rate. The variation of the learning rate is shown in Fig. 11(a). These parameters are determined based on many times of repetitive experiments on the training images.

In the training process, the weights of the feature extraction network in Fig. 3 are initialized with the pretrained classification model [33], [38], which can be viewed as transfer learning. Fig. 11(b) shows the variation of total loss during the training. The red line shows the loss along with the times of iteration and the green line shows the envelope of it. It can be seen from Fig. 11 that the loss function converges to the local optimal value with the increase of training steps.

To understand the detection results of UPN directly, an image patch is chosen from testing images. This patch and corresponding detection results are shown in Fig. 12. Fig. 12(b) shows the extracted high-dimensional feature maps of the ASPP module in UPNs and these maps have discrepant system responses to the different regions of input patch Fig. 12(a), which means that the well-trained UPNs could extract powerful and representative features [51]. Fig. 12(c) and (d) shows the probability map and detection

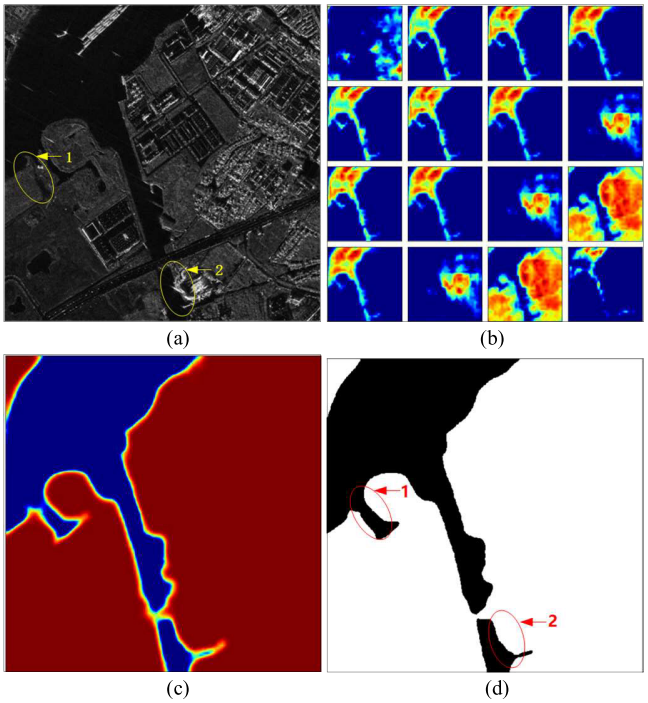

Fig. 12. Input SAR image patch and corresponding detection results by UPN. (a) Input image. (b) Output feature maps of ASPP. (c) Probability map. (d) Detection result.

result, and most of the regions are classified correctly. However, because of the resolution loss in UPNs, the boundaries of the land-water crossing regions in Fig. 12(a) (the two yellow ellipsoids) become ambiguous in Fig. 12(d) (the red areas).

2) Second Training Stage: In the second training stage, the proposed FC-CRF model is trained with SAR training images and the probability maps generated by UPNs. The mean-field approximation of FC-CRF is computed for five iterations. For the hyper-parameters of FC-CRF, the filter size k of variance-sharing convolution in (6) is chosen as 13 after a few trials. Variances of Gaussian filters θ_{d} are initialized by the grid-search [43] and updated by the back-propagation. The cross-entropy loss in (10) is used to train the FC-CRF model, for that this function ensures a much more precise segmentation boundary of coastal regions compared with the VFL function. The detection results with FC-CRF and without FC-CRF are shown in Fig. 13.

Fig. 13(a) shows the ground truth of Fig. 12(a) and Fig. 13(b) shows the detection result with FC-CRF. As shown in Fig. 13(b), the detection result of the proposed FC-CRF is very close to the ground truth. Particularly, Fig. 13(c) and (d) shows the difference images between ground truth and detection results without/with FC-CRF. In these difference images, the white pixels represent misclassified pixels and the black pixels represent correct-classified pixels. The fewer white pixels, the better the detection accuracy. As shown in Fig. 13(c) and (d), the proposed FC-CRF method could improve detection accuracy, especially more for the red region one and the yellow boundary 2. Therefore, combined with the well-trained UPNs and FC-CRFs, the proposed CFCNs

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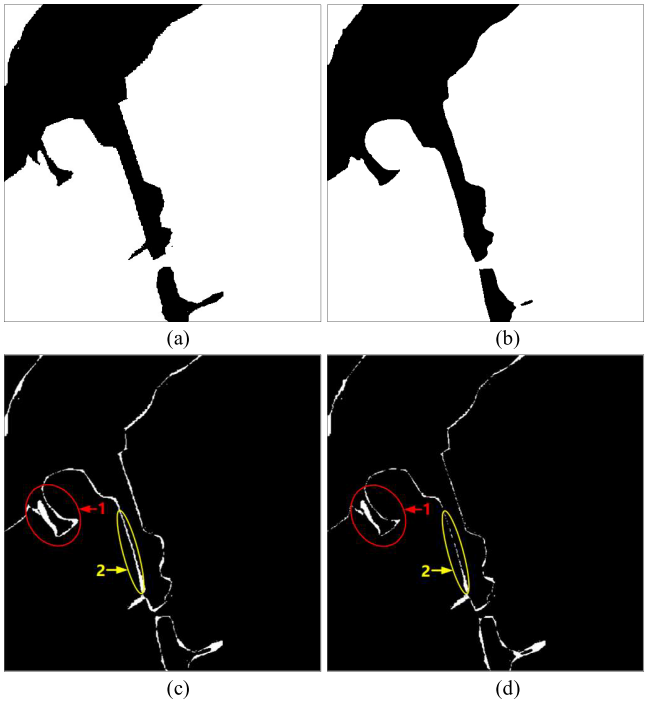

Fig. 13. Detection results comparison of UPN and FC-CRF. (a) Ground truth of Fig. 12(a). (b) Detection result with FC-CRF. (c) Difference image without FC-CRF. (d) Difference image with FC-CRF.

could realize the water body detection in high-resolution SAR images.

C. Performance Comparison of Different Methods

1) Evaluation Metrics: Mean pixel accuracy (MPA) and frequency weighted intersection over union (FWIoU) [52] are utilized in this article to evaluate the detection pixel accuracy quantitatively. n is the number of pixels' categories. p_{ii} is the truepositive pixels. p_{ij} and p_{ji} are the false-positive and false-negative pixels, respectively [33]. MPA is a metric in which the ratio of correct pixels is computed in a per-class basis and then averaged over the total number of classes

$$\mathrm{MPA} = \frac{\sum_{i=0}^{n} \mathrm{p}_{ii}}{\sum_{i=0}^{n} \sum_{j=0}^{n} \mathrm{p}_{ij}}. \tag{18}$$

Like the definition of MPA, FWIoU can be expressed as

$$\mathrm{FWIoU} = \frac{1}{\sum_{i=0}^{n} \sum_{j=0}^{n} \mathrm{p}_{ij}} \sum_{i=0}^{n} \frac{\sum_{j=0}^{k} \mathrm{p}_{ij} \mathrm{p}_{ii}}{\sum_{j=0}^{n} \mathrm{p}_{ij} + \sum_{j=0}^{n} \mathrm{p}_{ji} - \mathrm{p}_{ii}}. \tag{19}$$

FWIoU is an improved mean intersection over union (MIoU), which weights each class's importance depending on their appearance frequency. Due to the imbalanced category distribution in the detection data set, FWIoU is more appropriate for the evaluation of pixel accuracy compared with MIoU. Both MPA and FWIoU fall into $[0, 1]$ and the greater the numerical value, the higher the detection accuracy.

Besides the pixel accuracy of entire images, the accuracy of the water body boundary is much meaningful for the water body detection in SAR images. For a high-resolution SAR image I with m pixels, assuming r_{d} as the reference edge image of I which is acquired by edge detection of ground truth of I, p_{d} as the predicted edge image of I which is acquired by edge detection of detection result of I. The Pratt's figure of merit (PFOM) is introduced as

$$\mathrm{PFOM} = \frac{1}{\max\left(N_{\mathrm{r}}, N_{\mathrm{p}}\right)} \sum_{k=1}^{N_{\mathrm{p}}} \frac{1}{1 + \beta \mathrm{d}(k)^2} \tag{20}$$

where N_{r} is the number of edge points of reference edge image r_{d} and N_{p} is the number of edge points of the predicted edge image p_{d}. β is a constant which is usually set as $1/9$. $\mathrm{d}(k)$ is the Euclidean Distance between the edge point k in p_{d} and its nearest edge point in r_{d}. PFOM falls into $[0, 1]$ and 1 is the ideal value, which means that the boundaries of detection result are identical with that of corresponding ground truth. The execution time based on E5-2643 CPU and NVIDIA TitanXp GPU is also shown in this article.

2) Results Analysis of UPNs and VFL: The proposed CFCNs are composed of two novel FCNs: UPNs and FC-CRFs. To verify the effectiveness of these networks, respectively, the water body detection results of UPNs and VFL are first analyzed in this section. To evaluate the discrimination of our UPNs and VFL, other related approaches are utilized to compare with the proposed network. As for unsupervised detection methods, improved OTSU based on threshold segmentation [14], MRF [16] based on statistical information, and LevelSet [53] based on pixel gradients are used in this section. As for traditional supervised detection methods, the combination of GLCM+SVM [54] and Gabor+RF [23] are adopted, respectively. In the training of these two methods, the high-resolution SAR images in the training data set are first cropped into image patches with size 11×11, then GLCM features and Garbor features based on multiscale and multidirection kernels are extracted from these patches. Radial basis function (RBF) kernel is utilized in the SVM classifier and the hyperparameters are chosen by cross-validation and grid search. In the RF approach, the number of trees within the forest is searched from 200 to 500 [13]. Besides these traditional detection methods, Unet [27] and Deeplabv3+ [39], which are the most typical FCNs in semantic segmentation, are compared in this section.

To visually understand the detection results of different methods, a GaoFen-3 testing image is chosen. As Fig. 14(a) shows, this image was acquired on SL mode and HH polarization, which keep the same working parameters as training images. It can be seen from Fig. 14(b)–(d) that traditional unsupervised methods can effectively extract amplitude variations and contour features from the input image, and the water areas with low amplitude and continuous characteristics in this image are classified correctly. However, due to the lack of supervision information, these methods cannot discriminate other areas, which shows similar features to water areas, e.g., subway and farmlands. Compared with these unsupervised methods, Fig. 14(e) and (f) shows that GLCM+SVM and Gabor+SVM perform better detection accuracy, especially decrease the misclassified pixels in Fig. 14(b)–(d). Although these methods could make use of training images and

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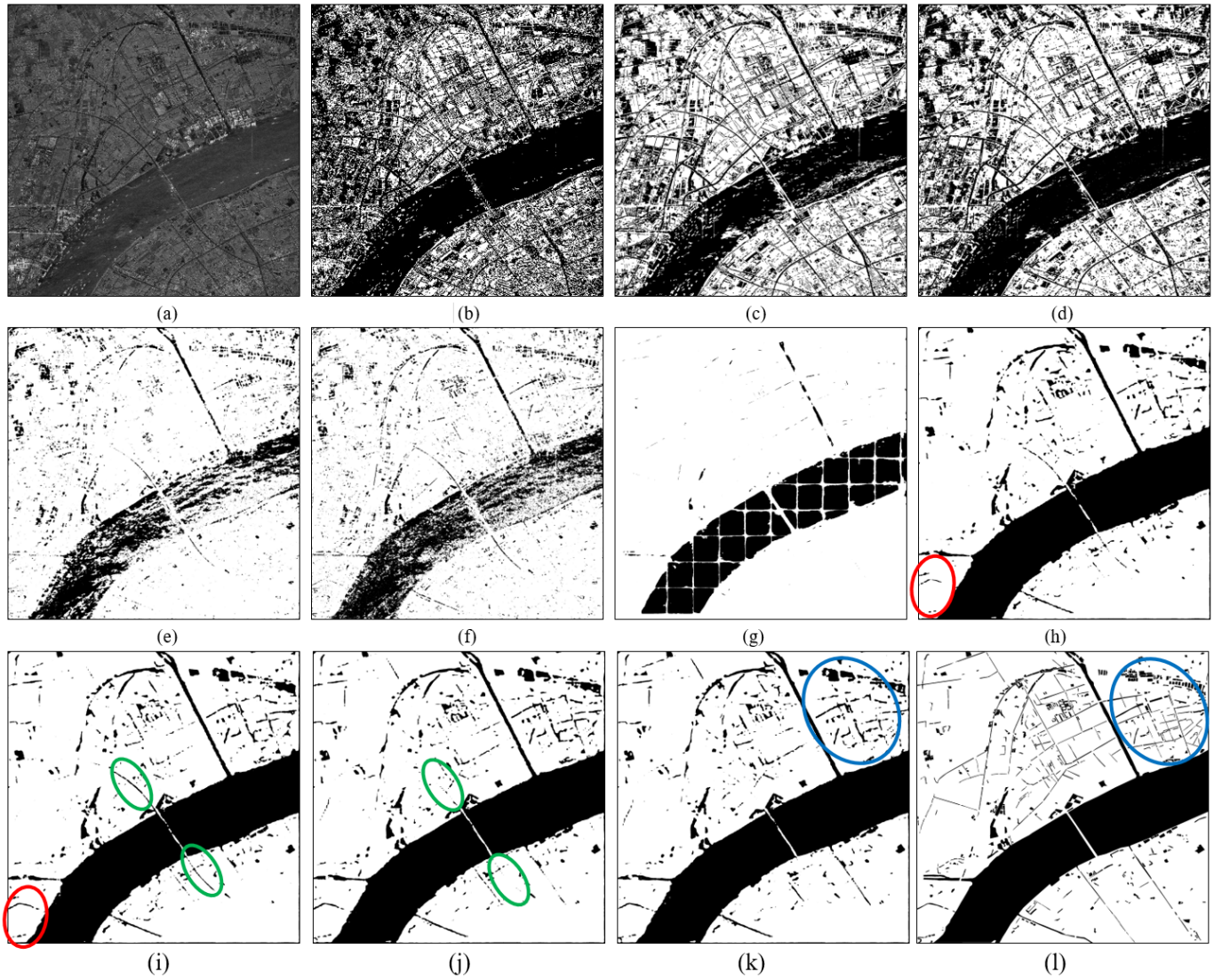

Fig. 14. Detection results of different methods. (a) SAR image acquired on SL mode and HH polarization, in Yangzhou, Jiangsu Province, China, with size 9555 × 9555 pixels and area 91.29 km^2. (b) MRF. (c) LevelSet. (d) OTSU. (e) GLCM+SVM. (f) Gabor+RF. (g) Unet. (h) Deeplab+CE. (i) DeepLab+VFL. (j) UPN+CE. (k) UPN+VFL. (l) Ground truth.

corresponding ground truths to extract powerful features and train effective classifiers, the detection results in Fig. 14(e) and (f) are still not as well as the ground truth in Fig. 14(l). Compared with the above traditional methods, deep learning methods perform better detection accuracy in Fig. 14(g)–(k). To be specific, Fig. 14(g) shows that Unet can detect water areas with a low false alarm. However, since that Unet only takes the image patches as input and it cannot deal boundary areas in each patch, there are a number of misclassified pixels of these boundaries in the detection result. Deeplabv3+, a popular detection network that can achieve very high accuracy in semantic segmentation, is also referred. The result of Deeplabv3+ in Fig. 14(h) shows that it can correctly classify most of the water areas, especially the water areas with isolated position and small size. To validate the effectiveness of the proposed VFL, we use this function to train a DeepLab network, and the result is shown in Fig. 14(i). Compared with Fig. 14(h) and (i), we can see that the introduction of the VFL function leads to more detailed detection results than CE, especially the red region. The reason lies in that the VFL function can adjust the loss contribution according to pixel frequency and pays more attention to difficult samples when training a network. The detection results of the proposed UPNs are shown in Fig. 14(j) and (k). As shown in these images, UPNs perform higher accuracy than Unet and Deeplab, especially for the highway in green regions which are wrongly classified as water areas in Fig. 14(i) has been correctly classified as land areas in Fig. 14(j). We attribute this result to the fact that UPNs can extract more powerful features with the combination of Xception and ASPP and establish a more reasonable decoder network with multiple upsampling layers. It is noticeable that the result in Fig. 14(k) is very close to the ground truth in Fig. 14(l) when introducing the VFL function to UPNs. Meanwhile, Fig. 14(k) shows that most kinds of water bodies, e.g., isolated ponds, lakes, and big rivers are correctly classified as water region, which performs that

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TABLE II
PERFORMANCE COMPARISON OF DIFFERENT METHODS

Method	MPA (%)	FWIoU (%)	PFOM (%)	Speed (s)
MRF [16]	63.03	61.08	15.98	112.55
LevelSet [19]	67.10	56.37	8.18	113.22
OTSU [14]	71.34	68.69	12.85	**1.69**
GLCM+SVM [54]	86.25	83.19	11.60	16042.31
Gabor+RF [23]	88.10	85.34	14.06	21754.63
Unet [27]	93.33	92.59	6.72	17.62
DeepLab+CE [39]	98.12	98.06	35.08	24.69
DeepLab+VFL	98.18	98.13	36.21	24.58
UPN+CE	98.39	98.48	39.73	26.13
UPN+VFL	**98.50**	**98.54**	**40.46**	26.20

the proposed UPN and VFL can mine the feature differences between different land backgrounds and water bodies and suppress the problem of performance degeneration caused by various water bodies.

Therefore, the visual results show that the combination of proposed UPNs and VFL function performs the highest detection accuracy for the testing image than other traditional detection methods and other popular deep networks.

Besides the above visual detection results, the quantitative results of Fig. 14 are also shown in Table II. As Table II shows, the water body detection methods based on deep learning perform better results than traditional unsupervised and supervised methods in detection pixel accuracy and boundary accuracy. The reason mainly lies in that the DNNs can make use of training images to extract more representative features by a number of convolution layers. For different deep learning methods, DeepLabv3+ performs higher detection accuracy than Unet for its more complicated network structure. It is noticeable that both pixel accuracy and boundary accuracy can be improved by introducing VFL to Deeplab. Especially, the UPN raises MPA and FWIoU to 98.39% and 98.48%, respectively. The combination of UPN and VFL performs the best MPA, FWIoU, and PFOM, which demonstrate the superiority of these proposed methods. What is important is that UPN can also run an inference with fast speed on GPU (nearly 26 s for a high-resolution SAR image with size 10 k $\times$ 10 k pixels).

3) Results Analysis of FC-CRF: As shown in Fig. 14, the proposed UPNs and VFL perform the best pixel accuracy in the testing image. However, due to the resolution loss in UPNs, the water boundaries of detection results are not as clear as the ground truth, especially for the blue region in Fig. 14(k). To validate the effectiveness of FC-CRF which is proposed to improve the water body boundary accuracy, the well-trained FC-CRFs are evaluated alongside the Grid-CRF of Shotton *et al.* [55], the FullCRF of Krähenbühl and Koltun [56] and the CRF-RNN of Zheng *et al.* [43]. To ensure a fair comparison, all the above models take the probability maps generated by UPNs+VFL as the unary potentials. The visual detection results of different CRFs are shown in Fig. 15. To visually understand the results of proposed FC-CRFs, the top right part of the SAR image in Fig. 15(a) is taken as the testing image for that this part has a number of narrow and isolated water areas, which reflect more boundary information than other areas. Fig. 15(c) shows the probability map generated by UPN+VFL, which is taken as the unary potentials of CRFs. Fig. 15(d) is the ground truth of cropped image Fig. 15(b). The detection results of different CRFs are shown in Fig. 15(e)–(h). It can be seen that different CRFs can suppress the resolution loss caused by UPNs with different extents. Fig. 15(h) shows that the proposed FC-CRFs can significantly improve water boundary accuracy. The difference images between the detection results in Fig. 15(e)–(h) of different CRFs and ground truth in Fig. 15(d) are also described in Fig. 15(i)–(l), respectively. The number of misclassified pixels (white pixels) performs the lowest in Fig. 15(l), which demonstrates the effectiveness of the proposed FC-CRFs.

TABLE III
PERFORMANCE COMPARISONS OF DIFFERENT CRFS

Method	MPA (%)	FWIoU (%)	PFOM (%)	Speed (s)
Unary	98.50	98.54	44.46	**26.13**
Grid-CRF [55]	98.46	98.56	48.98	62.37
FullCRF [56]	98.53	98.63	50.13	33.29
CRF-RNN [43]	98.74	98.76	51.97	28.96
FC-CRF	**98.92**	**98.81**	**54.66**	27.02

The quantitative results of Fig. 15 are shown in Table III. As Table III shows, all CRFs lead to a few increases in MPA and FWIoU. What is essential is that the proposed FC-CRF can significantly improve the PFOM than other CRFs, which are consistent with the detection results in Fig. 15. It is noticeable that proposed FC-CRFs can run an inference with a little increment in running time. The reason mainly lies in that proposed FC-CRFs reformulate all critical steps in mean-field approximation, especially the Message Passing, as standard convolutional layers which realize the end-to-end optimization of CRF models.

D. Performance Comparison of Different Imaging Modes

Due to the difference of working parameters under different imaging modes (as shown in Table I), the backscattering characteristics of water areas and land areas will change, which will lead to the feature differences in acquired SAR images. In order to test the CFCN's adaptability to the variation of radar parameters, the well-trained CFCN, which is trained with ten high-resolution images acquired under SL mode and HH polarization, is evaluated on a few Gaofen-3 SAR images which are acquired under multiple working modes and multi-polarization.

It can be seen from Fig. 16 that the proposed CFCN achieves well detection performance under various working modes in different regions. To be specific, CFCN classifies the red region in Fig. 16(a), which is blurred by strong scattering energy of ship targets, as water areas correctly. Most of the bridges in Fig. 16(b) are classified as land areas correctly. Although the electromagnetic reflection energy of the airport runway (green region) in Fig. 16(c) is relatively weak, and the amplitude information in this area is weaker than the other land areas, CFCN can also classify this region as land areas correctly. Besides that, the large marine region in Fig. 16(c) is correctly classified as water body. Based on the above

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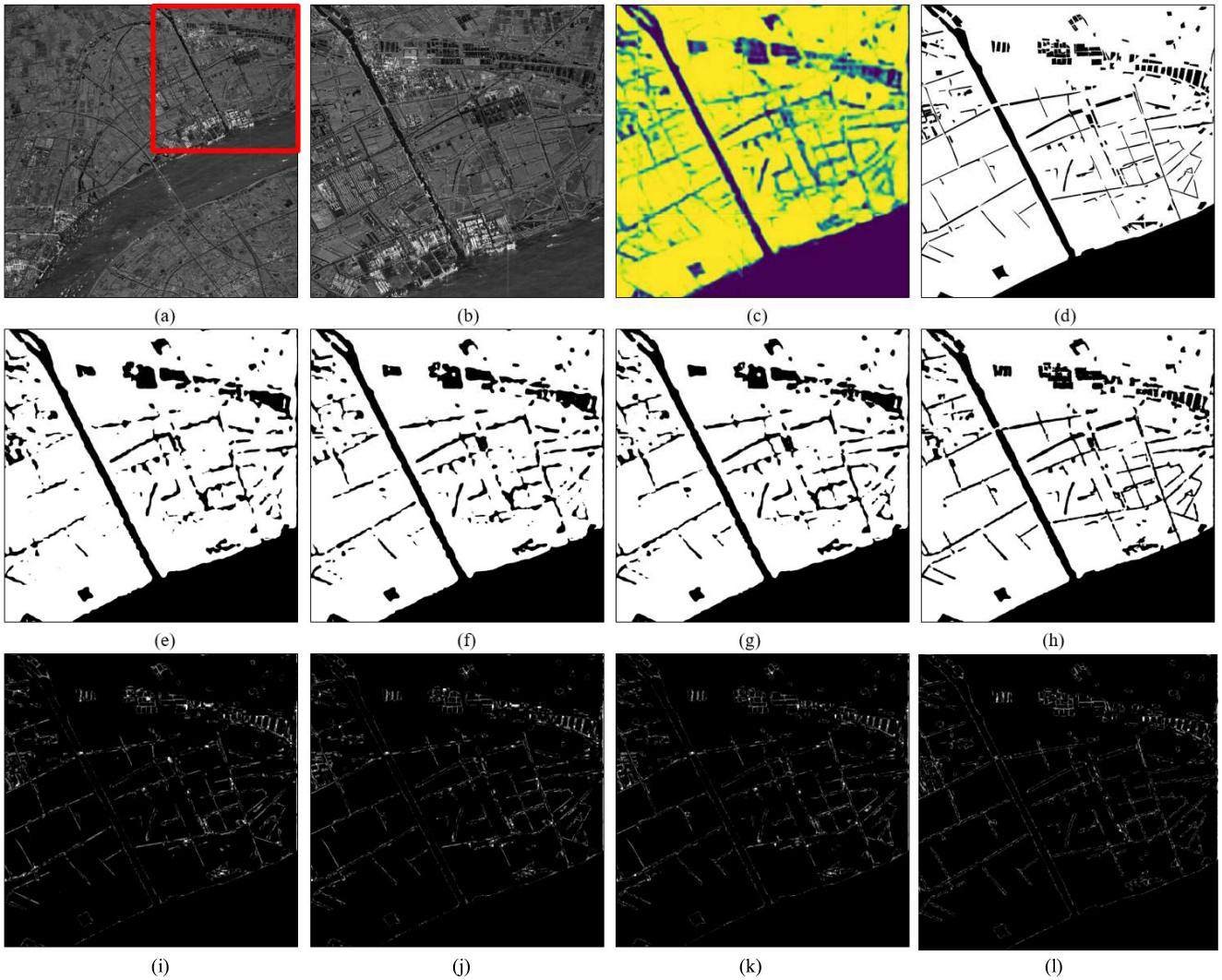

Fig. 15. Detection results of different CRFs. (a) Original SAR images. (b) Image patch of the red region in (a). (c) Probability map generated by UPN+VFL. (d) Ground truth. (e) Grid-CRF. (f) FullCRF. (g) CRF-RNN. (h) FC-CRF. (i) Difference image between (e) and (d). (j) Difference image between (f) and (d). (k) Difference image between (g) and (d). (l) Difference image between (l) and (d).

analysis, the proposed CFCN method has well adaptability and robustness to the various water bodies in high-resolution SAR images.

The working parameters and quantitative detection results of Fig. 16 are shown in Table IV. As Fig. 16 and Table IV show, these images acquired on different modes have different scene size, polarization modes, and resolution. Since the resolution differences and polarization differences between training images and testing images, the detection results of Fine Stripmap I (FSI) mode, Fine Stripmap II (FSII) mode, and Ultra-fine stripmap (UFS) mode are not as well as that of SL mode. Besides these differences, it can be seen that the proposed CFCN and VFL still perform well in pixel accuracy and boundary accuracy. The running time of CFCN varies linearly with the size change of the input SAR image. In particular, the time is 27.02 s for the image of $9k \times 9k$ pixels under SL mode and 36.80 s for the image of $12k \times 12k$ pixels under UFS mode, respectively. Therefore, the CFCN can achieve high performance and fast running speed under different imaging modes.

TABLE IV
PERFORMANCE COMPARISONS OF TESTING IMAGES ACQUIRED ON DIFFERENT IMAGING MODES AND DIFFERENT REGIONS

Imaging mode	Training	Testing			
	SL	SL	FSI	FSII	UFS
Size (pixels)	$9k \times 9k$	$9k \times 9k$	$10k \times 11k$	$6k \times 7k$	$12k \times 12k$
Polarization	HH	HH	HV	HV	HH
Resolution(m)	1.0	1.0	5.0	8.0	3.0
MPA (%)	—	98.92	94.89	95.16	96.32
FWIoU (%)	—	98.81	94.81	94.93	96.45
PFOM (%)	—	54.66	43.37	48.64	44.72
Speed (s)	—	27.02	32.07	25.41	36.08

E. Performance Comparison on Ship Detection

Ship detection from complex background SAR images is a challenging task due to the varying local clutter and low signal-to-clutter ratio. Especially, for the ship detection at land–water crossing regions, the complex land background with strong scattering energy will cause severe false alarm

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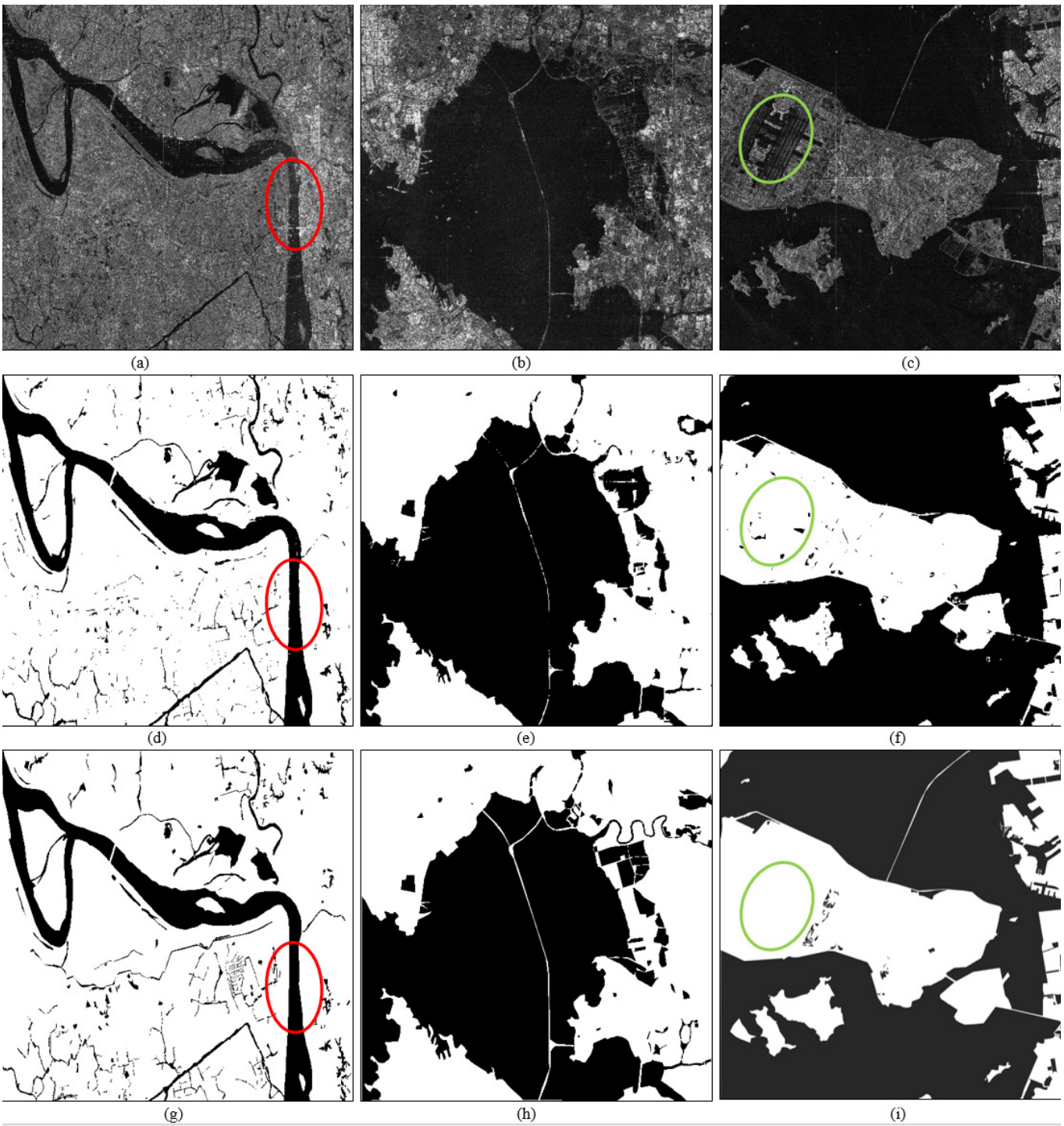

Fig. 16. Testing SAR images under different working modes and corresponding detection results by well-trained CFCN. (a) FSI mode in Anhui province, China. (b) FSII mode in Shandong province, China. (c) UFS mode in Inchon area, South Korea. (d) Detection result of (a). (e) Detection result of (b). (f) Detection results of (c). (g) Ground truth of (a). (h) Ground truth of (b). (i) Ground truth of (c).

targets in the detection results. Thus, the complex land background must be removed before detecting. The proposed CFCN and VFL function cannot only extract water body from high-resolution SAR images but also remove the complex land background at the same time, which means that these methods can also be used as a preprocessing step in the ship detection.

To validate the performance on ship detection of proposed methods, an SAR image patch is cropped from Fig. 16(a) and it is shown in Fig. 17(a). There are a number of ships in this image while most of them are surrounded by Yangtze River and some of them dock on the shore. The improved OTSU algorithm [14] is used as the comparison method for it performs well in land removal of SAR images. The

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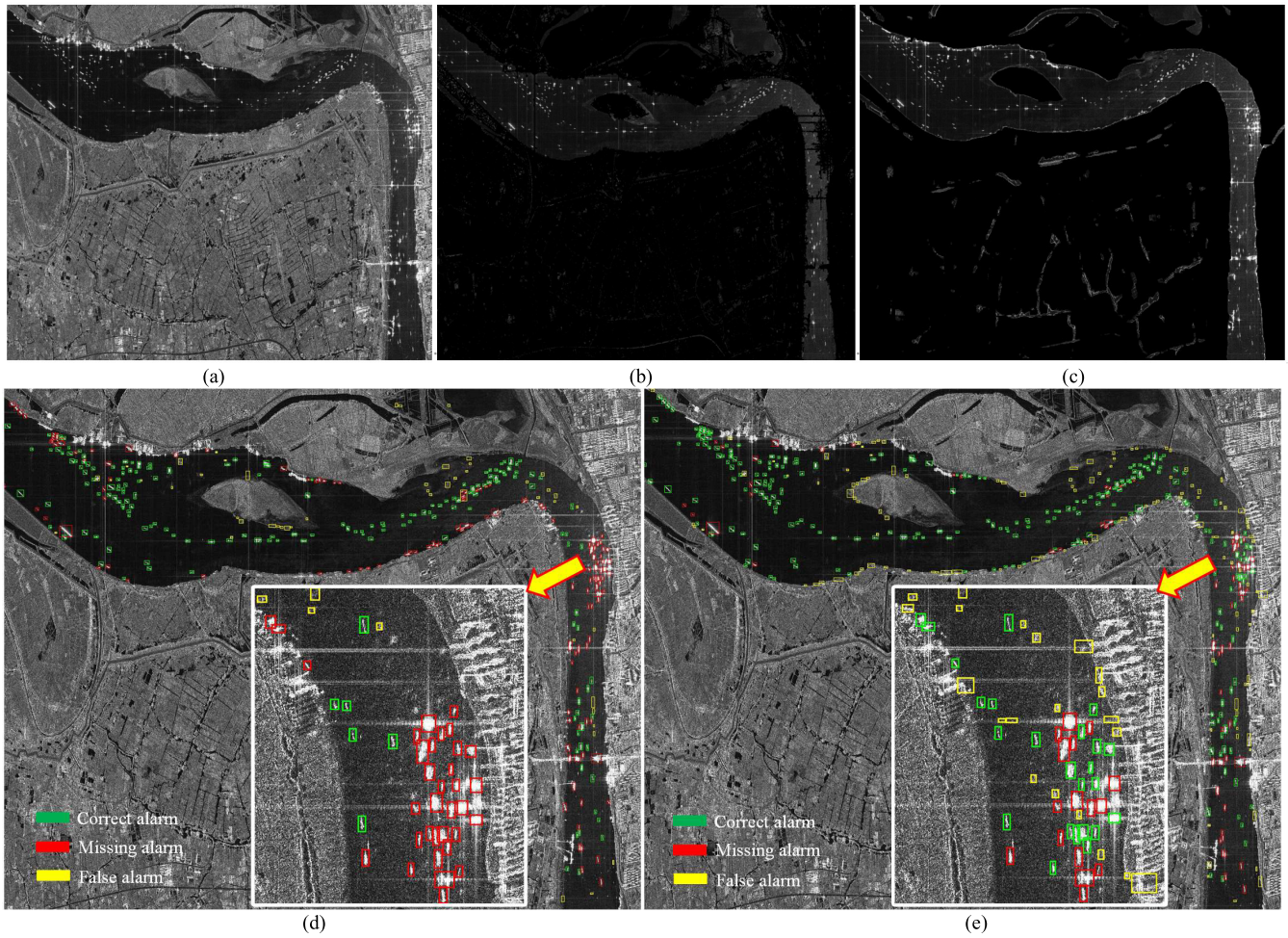

Fig. 17. Ship detection results of different methods. (a) SAR image patch cropped from Fig. 16(a). (b) Land removal result by the improved OTSU method. (c) Land removal result by the proposed CFCN+VFL method. (d) Ship detection result by the improved OTSU method. (e) Ship detection result by proposed CFCN+VFL method.

TABLE V
SHIP DETECTION PERFORMANCE OF DIFFERENT LAND REMOVAL METHODS

	Correct	Missing alarm	False alarm	Figure of merit
OTSU	169	125	46	49.70%
CFCN	231	63	122	55.52%

results of OTSU method and CFCN method are shown in Fig. 17(b) and (c), respectively. Although most land backgrounds are removed in Fig. 17(b), the coastline contour becomes ambiguous and some water areas containing ships are wrongly removed. The reason mainly lies in that the OTSU method is sensitive to the image intensity and noise cluttering. Meanwhile, strong scattering energy of ship targets near the coastline makes them easily classified as land regions. The result of Fig. 17(c) shows that the coastline contour and ships blurred by speckle noise are effectively preserved by the proposed CFCN method, which means that more ships can be extracted from this SAR image. To further analyze the ship detection performance, the CFAR method with global threshold [57] is used to detect the ships from the land removal images. The river area is set to obey Gaussian distribution and the false alarm probability is set 10^{-3}. The CFAR detection results of Fig. 17(b) and (c) are shown in Fig. 17(d) and (e), respectively, and one can see that most of the ships are correctly detected in these results. The differences between Fig. 17(d) and (e) mainly lie in that there are more missing alarm targets in Fig. 17(d) especially in the region with high noise level and there are more false alarm targets in Fig. 17(e) especially in the coastal areas. Therefore, the proposed CFCN methods perform better detection probability than OTSU method in the coastal region and noisy region. The quantitative results of Fig. 17 in Table V also show that the proposed CFCN method could significantly improve the correct alarm and reduce the missing alarm at the cost of increasing false alarm. Meanwhile, the CFCN method improved the figure of merit from 49.70% to 55.52%. Thus, the proposed method has much better adaptability than the other land removal methods under complex background cluttering, which indicates that it is an effective way to enhance ship detection.

V. CONCLUSION

In this article, the issue of water body detection in high-resolution SAR images has been discussed. The proposed CFCN integrated UPN with the FC-CRF model. The UPNs is proposed to realize the pixel-to-pixel water body detection. Then the FC-CRF model is introduced to realize the automatic

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learning of Gaussian convolution kernels and outputs a more accurate segmentation result. Furthermore, a novel VFL function is proposed to eliminate the inadequate training caused by the imbalanced data set. The experimental results on the multimode and multipolarization SAR images show that the proposed method can not only improve the pixel accuracy and boundary accuracy than traditional methods but also perform well in detection robustness and speed. Meanwhile, since the water body detection belongs to the two-class pixel-to-pixel classification, the proposed methods can not only be used in the water body detection, but can also be used in the landmark classification of high-resolution SAR images. Furthermore, for the ship detection in land-water crossing regions, the CFCN method could significantly improve the correct alarm and reduce the missing alarm at the cost of increasing false alarm under complex background cluttering, which indicates that it is an effective way in enhancing ship detection.

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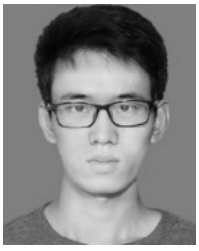

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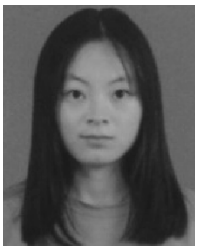

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